

Chat gpt :

Bhai, mast idea hai 🙌 A structured **18-day advanced level Qiskit roadmap** will be very impactful.

Since you want **theory + code + hands-on experiments** in Jupyter Notebooks, I'll design it like a mini-bootcamp where each day is a focused module building towards full mastery.

Here's a proposed **18-Day Quantum Computing with Qiskit (Advanced Level) Roadmap**:

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# 18-Day Quantum Computing with Qiskit – Advanced Roadmap

## Module 1: Advanced Quantum Circuits & Simulation (Day 1–3)

- **Day 1:** Advanced Qiskit Essentials
    - Qiskit 2.x overview (latest syntax changes)
    - Efficient circuit building (transpiler passes, parameterized circuits)
    - Hands-on: Build + transpile a multi-qubit circuit
  - **Day 2:** Quantum Simulation with Qiskit Aer
    - Noise models, custom backends
    - Statevector vs. density matrix simulator
    - Hands-on: Simulate decoherence on Bell states
  - **Day 3:** Circuit Optimization & Visualization
    - Gate fusion, depth reduction, CX count optimization
    - Hands-on: Optimize Grover's circuit on 5-qubit backend
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## Module 2: Quantum Algorithms (Day 4–8)

- **Day 4:** Grover's Algorithm Deep Dive
    - Amplitude amplification, oracle construction
    - Hands-on: Search problem with 3-qubit oracle
  - **Day 5:** Quantum Fourier Transform (QFT)
    - Theory, inverse QFT, applications
    - Hands-on: Build & benchmark QFT circuits
  - **Day 6:** Shor's Algorithm (Factoring)
    - Modular exponentiation, period finding
    - Hands-on: Simulate factoring of small numbers
  - **Day 7:** Quantum Phase Estimation (QPE)
    - Eigenvalue estimation, connection with QFT
    - Hands-on: Estimate phase of unitary operator
  - **Day 8:** Advanced Algorithm Projects
    - Combine QFT + QPE for chemistry applications
    - Hands-on: Phase estimation for H<sub>2</sub> molecule
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## Module 3: Variational Quantum Algorithms (Day 9–12)

- **Day 9:** Introduction to VQAs
    - Variational circuits, cost functions, optimizers
    - Hands-on: Build a simple VQE circuit
  - **Day 10:** VQE for Quantum Chemistry
    - Pauli decomposition, Hamiltonian simulation
    - Hands-on: Ground state energy of H<sub>2</sub>
  - **Day 11:** Quantum Approximate Optimization Algorithm (QAOA)
    - Combinatorial optimization problems
    - Hands-on: MaxCut problem with QAOA
  - **Day 12:** Advanced VQA Techniques
    - Hardware-efficient ansatz, barren plateaus
    - Hands-on: Compare optimizers & ansatz performance
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## Module 4: Quantum Machine Learning (Day 13–15)

- **Day 13:** Quantum Feature Maps & Kernels
  - Quantum data encoding, kernel methods
  - Hands-on: Quantum kernel SVM with Qiskit Machine Learning
- **Day 14:** Quantum Neural Networks (QNNs)
  - Parameterized circuits as neural layers
  - Hands-on: Train a QNN for classification
- **Day 15:** Hybrid Quantum-Classical ML

- TensorFlow/PyTorch integration with Qiskit
- Hands-on: Hybrid pipeline on real dataset

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## Module 5: Hardware Execution & Advanced Topics (Day 16–18)

- **Day 16: Running on Real Quantum Hardware**
  - IBM Quantum backends, job management, error mitigation
  - Hands-on: Execute VQE on 5-qubit IBM backend
- **Day 17: Error Mitigation & Quantum Error Correction (QEC)**
  - Zero noise extrapolation, readout error mitigation
  - Hands-on: Apply error mitigation on noisy circuits
- **Day 18: Capstone Project & Wrap-Up**
  - Choose a project:
    - Quantum Chemistry (VQE/QPE)
    - Optimization (QAOA)
    - Quantum ML Model
  - Present results with Jupyter Notebook + visualization

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## Repo Structure

```
18Day-Qiskit-Training/
├── README.md
├── requirements.txt
├── Day01_Advanced_Circuits.ipynb
├── Day02_Simulation.ipynb
├── ...
├── Day18_Capstone_Project.ipynb
├── resources/
│   ├── papers/
│   ├── cheat_sheets/
│   └── references.md
```

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👉 This roadmap gives **theory** → **code** → **hands-on** every day, progressing from circuits → algorithms → VQAs → ML → hardware. By **Day 18**, learners will be able to run real projects and even contribute to research.

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Bhai, do you want me to also **draft a README.md template for the GitHub repo** (with description, installation steps, contribution guidelines, etc.) so your repo is fully professional and ready for others to use?

### Gemini :

Learning Qiskit from a beginner to an advanced level requires a structured roadmap that builds from foundational concepts to complex applications. This 18-day plan provides a structured approach, starting with the basics and progressing to more advanced topics.

### Phase 1: Foundations (Days 1-6)

This phase is all about understanding the core principles of quantum mechanics and how they are applied in quantum computing. You'll get familiar with the fundamental building blocks before diving into code.

- **Day 1: Introduction to Quantum Computing.** Learn about the differences between classical bits and **quantum bits (qubits)**. Understand the key concepts of **superposition** and **entanglement**.
- **Day 2: The Bloch Sphere and Qubit States.** Visualize qubit states using the **Bloch sphere**. Learn about the mathematical representation of a single qubit, including **ket notation ( $|\psi\rangle$ )** and **complex numbers**.
- **Day 3: Single-Qubit Gates.** Explore the basic quantum gates that manipulate a single qubit, such as the **Pauli-X, Y, and Z gates**, and the **Hadamard gate**. Learn how these gates correspond to rotations on the Bloch sphere.
- **Day 4: Multi-Qubit Systems and Entanglement.** Understand how to represent systems with multiple qubits. Dive deeper into the concept of entanglement and its importance in quantum computing.
- **Day 5: Qiskit Installation and First Circuit.** Set up your Python environment and install Qiskit. Write your first "Hello World" quantum circuit using Qiskit to create and measure a qubit.
- **Day 6: Simulating and Running Circuits.** Learn to use Qiskit's simulators, like **Qiskit Aer**, to test your circuits. Run your first circuit on a real IBM Quantum computer via the cloud platform.

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### Phase 2: Intermediate Algorithms (Days 7-12)

Now that you have the basics down, this phase focuses on implementing and understanding key quantum algorithms that demonstrate a computational advantage.

- **Day 7: Deutsch-Jozsa Algorithm.** Study the Deutsch-Jozsa algorithm, a classic example of how a quantum computer can solve a problem faster than a classical computer. Implement it in Qiskit.
- **Day 8: Bernstein-Vazirani Algorithm.** Explore another foundational quantum algorithm. Learn how it finds a hidden string with a single query, an exponential speed-up over classical methods.

- **Day 9: Grover's Search Algorithm.** Dive into Grover's algorithm, a powerful search algorithm that provides a quadratic speed-up for unstructured search problems. Implement the algorithm for a small-scale problem.
- **Day 10: Quantum Teleportation and Superdense Coding.** Understand how quantum protocols use entanglement to transmit information. Implement both quantum teleportation and superdense coding in Qiskit.
- **Day 11: Quantum Fourier Transform (QFT).** Learn the QFT and its role in many advanced quantum algorithms. Implement a basic QFT circuit.
- **Day 12: Quantum Phase Estimation (QPE).** Study QPE, an algorithm used to estimate the eigenvalues of a unitary operator. This is a crucial subroutine for more complex algorithms like Shor's algorithm.

### Phase 3: Advanced Concepts and Applications (Days 13-18)

In this final phase, you'll tackle more complex topics, including near-term applications and more advanced features of Qiskit.

- **Day 13: Variational Quantum Eigensolver (VQE).** Learn about hybrid quantum-classical algorithms. Focus on VQE, a method used in quantum chemistry to find the ground state energy of a molecule.
- **Day 14: Quantum Approximate Optimization Algorithm (QAOA).** Study QAOA, another hybrid algorithm for solving combinatorial optimization problems. Implement a small example.
- **Day 15: Error Mitigation and Correction.** Understand the challenges of noise in quantum computers (**decoherence**). Learn about techniques like **error mitigation** to improve the accuracy of results on current hardware.
- **Day 16: Qiskit Primitives.** Explore the new Qiskit Primitives, such as **Sampler and Estimator**, which simplify the process of running quantum programs on real hardware.
- **Day 17: Qiskit Machine Learning.** Discover the emerging field of **quantum machine learning**. Implement a simple quantum classification model using Qiskit.
- **Day 18: Beyond the Basics.** Research more advanced topics like dynamic circuits, Qiskit's transpilation services, or quantum error-correcting codes. Explore the Qiskit documentation and community for further learning.

By following this roadmap, you'll gain a solid theoretical understanding of quantum computing and the practical skills to implement and experiment with quantum circuits using Qiskit.

You can find a great overview of the latest features and how to prepare for a certification exam in this video: Qiskit Developer Certification 2.0: How to Prepare.

Copilat :

Got it  — you want an **18-day, advanced-level, hands-on Qiskit training roadmap** that's structured, modular, and packed with theory + code + experiments. Here's a **day-by-day progression** that builds depth while keeping it practical for someone already comfortable with quantum basics.

## 18-Day Advanced Qiskit Roadmap

**Format:** Each day = 1 Jupyter Notebook with

- **Theory** (concise, advanced-level)
- **Code** (Qiskit implementation)
- **Hands-on Experiment** (simulator + hardware where possible)
- **Challenge Task** (push beyond the tutorial)

### Phase 1 — Advanced Foundations (Days 1-4)

| Day | Focus                             | Key Outcomes                                                                                               |
|-----|-----------------------------------|------------------------------------------------------------------------------------------------------------|
| 1   | Advanced Qiskit Setup & Runtime   | Optimize environment, use Qiskit Runtime, manage backends, job monitoring, transpiler optimization levels. |
| 2   | Mathematical Backbone             | Tensor products, unitary decomposition, Kraus operators, density matrices.                                 |
| 3   | Advanced Gates & Custom Unitaries | Implement multi-controlled gates, arbitrary rotations, custom gate definitions.                            |
| 4   | Noise Models & Error Channels     | Build custom noise models, simulate decoherence, amplitude/phase damping, depolarizing channels.           |

### Phase 2 — Core Quantum Algorithms (Days 5-10)

| Day | Focus                                             | Key Outcomes                                                   |
|-----|---------------------------------------------------|----------------------------------------------------------------|
| 5   | Quantum Fourier Transform (QFT)                   | Implement QFT & inverse QFT, apply to period finding.          |
| 6   | Quantum Phase Estimation (QPE)                    | Apply QPE to eigenvalue problems & factoring subroutines.      |
| 7   | Grover's Algorithm – Advanced                     | Multi-solution search, oracle design, amplitude amplification. |
| 8   | Shor's Algorithm                                  | Full factoring pipeline with modular exponentiation circuits.  |
| 9   | Variational Quantum Eigensolver (VQE)             | Chemistry Hamiltonians, ansatz design, optimizer tuning.       |
| 10  | Quantum Approximate Optimization Algorithm (QAOA) | Graph problems, parameter landscapes, performance scaling.     |

### Phase 3 — Specialized Applications (Days 11-14)

| Day | Focus                           | Key Outcomes                                                  |
|-----|---------------------------------|---------------------------------------------------------------|
| 11  | Quantum Machine Learning (QSVM) | Implement QSVM for classification tasks, kernel optimization. |

| Day | Focus                                        | Key Outcomes                                                  |
|-----|----------------------------------------------|---------------------------------------------------------------|
| 12  | Hybrid Quantum Neural Networks (QNNs)        | TorchConnector, hybrid training loops, gradient analysis.     |
| 13  | Quantum Simulation of Physics                | Trotterization, simulating spin chains, Hubbard model basics. |
| 14  | Quantum Cryptography & Post-Quantum Security | BB84, E91 protocols, quantum key distribution simulation.     |

Phase 4 — Hardware & Scaling (Days 15-18)

| Day | Focus                             | Key Outcomes                                                                                        |
|-----|-----------------------------------|-----------------------------------------------------------------------------------------------------|
| 15  | Pulse-Level Programming           | Qiskit Pulse, custom schedules, Hamiltonian calibration.                                            |
| 16  | Error Mitigation & Correction     | Zero-noise extrapolation, measurement error mitigation, surface code basics.                        |
| 17  | Multi-Backend & Cloud Integration | Running jobs across IBM, AWS Braket, and hybrid workflows.                                          |
| 18  | Capstone Project                  | Choose a real-world problem → design, implement, and benchmark a quantum solution. Present results. |

📌 Extra Tips for Impact

- Repo Structure:
- `/Day01_AdvancedSetup.ipynb`
- `/Day02_MathBackbone.ipynb`
- `...`
- `/Day18_Capstone.ipynb`
- `README.md` (with roadmap + install guide)
- `/resources` (papers, diagrams, datasets)
- Challenges:** End each notebook with an “Open Problem” to encourage exploration.
- Visualization:** Use Bloch spheres, circuit drawers, and matplotlib for clarity.
- Hardware Runs:** Schedule at least 3–4 real backend executions to show noise effects.

If you want, I can also **design the README + visual roadmap graphic** so your GitHub repo instantly communicates the 18-day journey and attracts stars 🌟. Do you want me to go ahead and **draft that README + visual flow** for you next? That would make your project lau

# Starting For Here,,

18 day Maha-Quantum War  
( Inspiratin for Mahabhart)

Module 0 :

Our\_Great\_warrier\_Intro:

1. Mathematical:
  - a. Basic Linear Algebra
  - b. Complex Number and probability
  - c. Classical Comutating Fundamental
  - d. Basic Computer Science
2. Intro to Quantum Concepts
  - a. Quantum Mechanics

Module 1 :

1. Qiskit 2.x overview (latest syntax changes)
2. Advanced Gates & Custom Unitaries: Implement multi controlled gates, arbitrary rotations, custom gate definitions.
3. Advanced Qiskit Essentials
  - a. Qiskit 2.x overview (latest syntax changes)
  - b. Efficient circuit building (transpiler passes, parameterized circuits)
  - c. Hands-on: Build + transpile a multi-qubit circuit
4. **Multi-Qubit Systems and Entanglement.** Understand how to represent systems with multiple qubits. Dive deeper into the concept of entanglement and its importance in quantum computing.
- 5.

Module 2 :

- **Day 2:** Quantum Simulation with Qiskit Aer
    - Noise models, custom backends
    - Statevector vs. density matrix simulator
    - Hands-on: Simulate decoherence on Bell states
- 1.